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ARTICLE INFO

Article history:

Received 19 December 2016

Revised 14 June 2017

Accepted 17 June 2017

Available online 10 December 2018

JEL classification:

G23

G32

Keywords:

Institutional ownership

Governance

Coordination

Exit

Voice

ABSTRACT

We examine the impact of investor coordination on governance. We identify coordinating groups of investors (cliques) as those connected through the network of institutional holdings. Clique members vote together on proxy items: a one standard deviation increase in clique ownership more than doubles votes against low quality management proposals. We use the 2003 mutual fund trading scandal to show that this effect is causal. These findings suggest coordination strengthens governance via voice. Coordination, however, also weakens governance via threat of exit. Clique owners exit positions more slowly, and firm value responds negatively to liquidity shocks when clique ownership is high.

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1. Introduction

If investors act independently, then ownership dispersion will impede governance, all else equal, because owners of small positions do not have the incentive to monitor individually (Shleifer and Vishny, 1986). However,

recent evidence suggests that some investors do not act independently, but instead work together to influence corporate policies in the firms they own (Edmans and Holderness, 2016). Anecdotal evidence also supports the view that owners work in concert to affect firm outcomes.¹ For the purposes of governance, such coordinated groups can effectively act as single blocks of ownership, even when each investor holds only a small stake.

In this paper, we examine the relationship between ownership structure and firm governance, taking into account investor interactions. We identify groups of investors that are likely to be working together to influence the firms they own. We then examine how the presence of these coordinating owners relates to governance. Shareholder coordination increases governance via voice by overcoming the free rider problem, consistent with Shleifer and Vishny (1986). At the same time, coordination

[☆] We thank Bill Schwert, the editor, and an anonymous referee for very helpful comments. We also thank Pat Akey, Alon Brav, Kevin Crotty, David De Angelis, Dave Denis, Alex Edmans, Joseph Gerakos, Stu Gillan, Todd Gormley, Jiekun Huang, Peter Iliev, Melissa Prado, Tao Shu, Chester Spatt, Jason Sturgess, Jing Zeng, and seminar participants at Carnegie Mellon, DePaul University (Economics), University of Georgia, Hanken School of Economics, University of Houston, University of Kentucky, University of Illinois Chicago, University of Nebraska, University of Pittsburgh, Texas Tech University, Ohio University, and participants at the 2016 LBS European Winter Finance conference, the 2016 European Finance Association conference, the 2016 CMU-Pitt-PSU Finance conference, the 2017 FSU Suntrust Beach conference, and the 2017 Drexel Corporate Governance conference for helpful comments.

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¹ See, for example, Foley and McLannahan (2016), Das (2015), and Benoit and Grind (2015).

weakens governance via threat of exit, as predicted in [Edmans and Manso \(2011\)](#).

Recent papers on investor interactions generally focus on observable events, such as activism campaigns, where coordination can be identified ex post (e.g., [Dimson et al., 2015](#); [Artiga González and Calluzzo, 2016](#); [Appel et al., 2016](#)). However, if coordination improves governance through the threat of exit or voice, then there may be no need for activist campaigns. To identify investor coordination ex ante, we turn to recent work at the intersection of network theory and economics. A standard finding in this literature is that networks of highly clustered communities (cliques) support coordination (see e.g., [Assenza et al., 2008](#); [Marcoux and Lusseau, 2013](#)). We apply this idea to the network of institutional investors. We also rely on insights from prior research that suggests that overlapping portfolio positions are correlated with interactions between investors ([Shiller and Pound, 1989](#); [Hong et al., 2005](#); [Pool et al., 2015](#)). Therefore, we define two institutions in our network to be connected if they each have a large ownership stake ($\geq 5\%$) in a given firm. A clique is a group of institutions in which every member is connected, through any stock, to every other member. For example, we estimate that well-known activists Carl Icahn and Larry Robbins (Glenview Capital Management) are in a clique together in 2013. At the end of 2012, Icahn and Robbins both owned a block in Take Two Interactive Software, Inc. They are also in the clique with Capital Research Global Investors, among others. Icahn and Capital Research were connected via a block in Netflix, and Robbins and Capital Research had a common block in Rovi Corporation. When Icahn engaged Apple in 2013, he only owned 0.5% of the firm; however, his clique owned 4.9%.

We identify roughly 20 institutional investor cliques each year. The median number of clique members is 24. Approximately 35% of all institutional investors belong to a clique in a given year. Coordinated groups, in total, own close to 30% of the average firm, and the single clique with the largest stake owns 13%.

We find that institutional ownership has become less concentrated over the past 30 years. The median stake of institutional owners in firms in 2013, 0.029%, is roughly five times smaller than it was in 1980 (0.128%), and ownership by the five largest institutions, as a fraction of total institutional ownership, has dropped from 83% to 60%.² All else equal, governance problems associated with dispersed ownership should be worse today than they were in 1980. However, this does not account for the possibility that institutions work together. In fact, total clique ownership in US firms has increased from about 12% in 1980 to 42% in 2013, suggesting that coordination is substituting for concentrated holdings.

To examine the relation between ownership by cliques and governance by voice, we study shareholder voting on management proposals, as voting is a direct form of investor engagement ([McCahery et al., 2016](#)). We expect

clique members to vote together, and in particular, against proposals that are not in shareholders' interests. Consistent with this, the votes of institutions in cliques are 8–10% more correlated with members of their clique than with other institutions. We find that, all else equal, a one standard deviation increase in total clique ownership increases votes against bad management proposals by 6.2 percentage points, relative to an unconditional average of 5.8%. This is robust to controlling for other characteristics of ownership structure, including the level of institutional ownership, ownership by blockholders, ownership concentration, and ownership by different types of institutions.

It is possible that cliques self-select into firms with ownership that systematically opposes bad management proposals, rather than improve the governance of the firms they own. To address this, we use the 2003 mutual fund scandal as a shock that provides plausibly exogenous variation in clique ownership. In September 2003, 25 mutual fund families were accused of illegal trading practices by the Securities and Exchange Commission ('SEC'). This scandal resulted in significant outflows from and closures of these institutions that impacted connections within the network of institutional investors and reshaped existing cliques. These changes in cliques lead to relevant changes in ownership structure at the firm level, and we argue that the scandal only impacts voting through its effect on ownership structure.³ Our results are robust to using this instrument. These findings suggest that there is a significant amount of coordination among investors and that this has meaningful effects on governance via voice.

Anecdotal evidence suggests that some institutions prefer specific corporate policies that they perceive as value enhancing, rather than broadly improving governance across all dimensions.⁴ We examine the relation between policy outcomes and clique ownership when cliques reveal a preference for a particular policy. We focus on four outcomes that are associated with investor activism and for which outcomes are easily observable: dividend initiations, share repurchases, divestitures, and the prevention of acquisitions. For each institution, we measure the fraction of the firms in its portfolio, that, for example, initiated a dividend in the prior year. We then take this value for the median institution in each clique to characterize the extent to which that clique exhibits a preference for that particular policy. We find that ownership by cliques with strong policy preferences predicts future policy outcomes in the firms they own. For example, a one standard deviation increase in dividend clique ownership is associated with a 4% increase in the probability of dividend initiation in the following year. These associations can reflect cliques forecasting future policy changes or causing those changes.

³ This shock is used to provide plausibly exogenous variation to common ownership by [Anton and Polk \(2014\)](#) and [Koch et al. \(2016\)](#), for example. We discuss in more detail the validity of this instrument in our setting in [Section 4.1.1](#).

⁴ For example, Icahn Capital LP often focuses its activism on pay-out policies. Moreover, institutions that attend the Shareholder's Rights Project clinic may have a particular goal of declassifying boards. Or similarly, the Council for Institutional Investors' most recent campaign focuses on majority versus plurality voting standards.

² [Ben-David et al. \(2016\)](#) show that the largest institutions own a larger fraction of the market than in the past. However, these institutions own a much larger number of firms as well. This has led to an overall decrease in firm level ownership concentration, on average.

Both interpretations are of interest, as each suggests that ownership by certain cliques predicts future firm policies.

As an alternative to direct intervention, investors can decide to sell their shares (exit) if dissatisfied with firms' policies, the threat of which acts as a governance mechanism (Admati and Pfleiderer, 2009; Edmans, 2009). Edmans and Manso (2011) show theoretically that coordination among owners weakens the threat of exit. This is because the threat of exit is strongest when owners trade independently and aggressively, impounding their information into the price. Multiple investors that can credibly commit to cooperate should trade less aggressively to minimize price impact than investors who cannot cooperate (Holden and Subrahmanyam, 1992; Foster and Viswanathan, 1996).

We test whether institutions' trades are a function of coordinated ownership. If cliques are measuring coordination, institutions in cliques should reduce their positions slowly over time to minimize price impact, rather than selling all at once. Consistent with this, trades are 6% more positively autocorrelated when institutions sell stocks they own with other clique members compared to those where they are the only clique member in the stock.

Identifying the effects of clique ownership on the threat of exit is challenging because the threat is not observable. We follow prior literature that links the threat of exit to liquidity (e.g., Edmans, 2009). Liquidity is good for governance via threat of exit (the sensitivity of firm value to liquidity is high) when large, multiple, independent investors own the firm. In contrast, we expect this effect to be weaker (the sensitivity of firm value to liquidity is low) when investors coordinate. Following Bharath et al. (2013) and Edmans et al. (2013), we measure the effect of decimalization (a shock to liquidity) on q to proxy for the threat of exit, and we interact this with clique ownership. We find that the threat of exit is weaker when clique ownership is high, even after controlling for ownership structure and other firm characteristics. The economic effect is large: the increase in q after decimalization is 0.5 lower if clique ownership is one standard deviation higher.

Coffee (1991) and Bhidé (1993) argue that liquidity can hinder governance via voice. Our tests of the relation between q and ownership around liquidity shocks could therefore be confounded by this alternative effect of liquidity. Therefore, we also examine the relation between clique ownership and managers' short-term concerns. Edmans (2009) argues that the threat of exit is particularly effective in firms with managers that are sensitive to the short-term stock price. If clique ownership hurts the threat of exit, then we expect cliques to avoid these firms. We proxy for managers' short-term concerns using the sensitivity of the value of vesting equity to the stock price over the next year. In addition, we use the CEO's age, as managers closer to retirement are more likely to make choices as a function of their short horizon (Dechow and Sloan, 1991; Antia et al., 2010). We find clique ownership is lower when managers have short-term concerns. For example, a one standard deviation increase in the sensitivity of vesting equity is associated with a one percentage point decrease in clique ownership.

Overall, our results are consistent with coordination among institutional investors facilitating governance by voice and mitigating governance by exit. One problem that researchers face when examining coordination is that it is difficult to distinguish coordination from agents acting independently but with similar information or preferences (Manski, 1993). This alternative explanation is inconsistent with our primary results. Informed agents acting independently should govern effectively through both voice and exit. In contrast, we find that the threat of exit is weaker. Furthermore, we find institutions in cliques are 40% more likely to disclose explicit joint activism with other investors, as measured by joint Schedule 13D form filings.⁵ We discuss this alternative explanation and present additional tests to address the concern in Section 6.

Our paper is related to, but distinct, from existing work on coordination and governance. There is evidence that formal shareholder organizations, such as Institutional Shareholder Services (ISS) and United Shareholders Association (USA), impact governance (e.g., Gillan and Starks, 2000; Bethel and Gillan, 2002; Doidge et al., 2015). We provide evidence of coordination beyond these formal organizations.⁶ Huang (2014) finds that firm value is higher when the firm's owners are geographically proximate or have highly correlated portfolios. Several papers examine activist campaigns. Artiga González and Calluzzo (2016) find that campaign success is positively correlated with the number of activists. Dimson et al. (2015) find that collaboration among stakeholders increases the chances of successful social responsibility activism. Appel et al. (2016) also examine activist campaigns and find that the presence of passive investors does not alter campaign frequency, but does change the type of activism.

Our conclusions differ from this prior research. We believe we are the first to show a trade-off in the governance effects of coordination; it improves voice but weakens exit. This trade-off has implications for policy makers. Regulations that restrict or ease shareholder communication continue to be a topic of debate. Our results suggest that, like many issues in governance, such regulations are not one size fits all. Restricting shareholder communication can improve governance in some firms while hurting governance in others.

We also contribute in our approach to identifying coordination. We allow investors to be connected through any ownership position, not just in the firm of interest. Our concept of coordination is therefore similar to that of multimarket contact in industrial organization in which firms collude more easily because they compete in many markets (Bernheim and Whinston, 1990). This concept can be applied to a variety of settings in which coordination is important, e.g., groups of analysts, mutual fund herding,

⁵ Schedule 13D forms are required when investors coordinate in certain activities and their joint ownership exceeds 5%.

⁶ There is evidence of coordination between firms as a result of having the same owner (e.g., Azar et al., 2018; Panayides and Thomas, 2016). This is distinct from coordination between owners in a given firm. Furthermore, Azar et al. (2018) find negative consequences of coordination (e.g., anticompetitive outcomes), whereas we find both positive and negative effects.

cliques of CEO's or board members, etc. In our setting, the approach allows us to consider investors with small stakes who engage the firm because of their connections to other owners. We can therefore abstract from the standard definition of block ownership based on a 5% ownership threshold. [Edmans and Holderness \(2016\)](#) note that this threshold is arbitrary, and consideration of smaller stakes can add to our understanding of relations between ownership structure and governance.

2. Identifying coordination in a network

An ideal setting to examine the impact of coordination on governance is one in which researchers perfectly observe all forms of coordination between investors in a large sample. Although some formal coordination mechanisms are observable (e.g., USA, ISS), informal coordination is difficult to observe. To overcome this, we proxy for the incentives and propensity of institutional owners to coordinate based on their connections within the network of US institutional investors. This proxy relies on the concept of complete subgraphs, also known in network theory as cliques.

Intuitively, a clique is a group of investors where all members are connected to one another. Theoretical and experimental work on coordination suggests that the survival of cooperative strategies is supported by the presence of these highly interconnected groups ([Santos and Pacheco, 2005](#); [Assenza et al., 2008](#); [Marcoux and Lusseau, 2013](#)). To identify these groups within the network of institutional investors, we define connections between institutions using their common holdings in the fourth quarter of each calendar year. Specifically, a connection exists between two investors if each owned a large stake (at least 5%) in at least one common firm at the end of the prior year. Our definition of a connection between two institutional investors assumes that sharing large common holdings in at least one firm increases the probability of interactions relative to institutions that do not share such common ownership. This assumption is supported by prior literature. [Hong et al. \(2005\)](#) and [Pool et al. \(2015\)](#) find that common holdings are associated with geographic proximity, and they conclude that this is likely driven by direct communication between shareholders that are physically near each other. [Shiller and Pound \(1989\)](#) survey managers and find that the most important driver of portfolio choices is direct discussions between investors. These findings are consistent with our assumption that common holdings are correlated with prior interactions between the institutions.⁷

The structure of connections between institutional investors is important for coordination. In [Fig. 1](#) below, investor A has the same connections in the examples given in both Panel (A) and Panel (B). However, investor A belongs to a clique (that includes all the other investors) only in Panel (A), despite the fact that investor A's connections are the same across the two examples. This makes intuitive sense as a proxy for coordination. Information

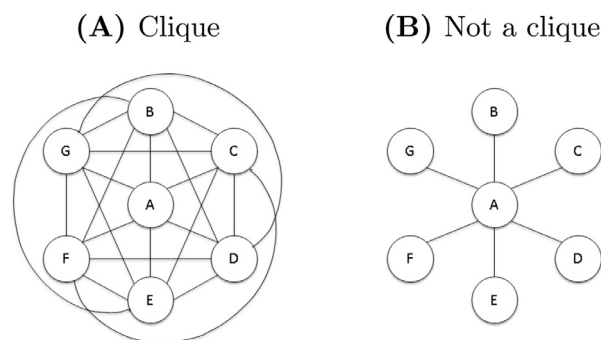


Fig. 1. Examples of subgraphs.

can move more easily from any investor in a clique to any other investor in Panel (A). In Panel (B), information must move through investor A, the central investor in the network, to other members. Coordination is therefore made more difficult by the lack of common holdings among the other investors in the network.

Each year, we construct an $N \times N$ matrix of institution-to-institution relationships. The off-diagonal elements take a value of one if both institutions in that pair have a position $\geq 5\%$ in common in at least one firm, and zero otherwise. We then estimate our measure of coordination from this entire network of overlapping ownership positions. This includes more information about institutional relationships relative to other measures that identify co-ownership only in the firm of interest (e.g., [Appel et al., 2016](#); [Artiga González and Calluzzo, 2016](#)). This is important because investors in a given clique are likely to be connected through a variety of different firms.

Exact identification of cliques in a network is a difficult computing problem that cannot be feasibly solved given the size and complexity of the network of institutional investors.⁸ However, network research has developed a variety of algorithms to approximate solutions to the problem of identifying cliques. We use the most recent and arguably best performing of these algorithms, the Louvain algorithm, developed in [Blondel et al. \(2008\)](#).

The Louvain algorithm groups investors such that the density of their connections within the group is high relative to the density of connections to investors outside of the group. The output is an assignment of each institution either to a specific clique or to no clique at all. The number and size of cliques are not input parameters, rather they are determined by the data. The algorithm is static in that each cross-section (based on reported holdings at the end of the year) is treated independently. Because exact clique identification is infeasible for a network this size, our measure is an approximation.

After identifying the institutions that belong to each clique in each year, we compute firm-level measures of ownership by cliques. We calculate three measures:

⁷ Interactions are not necessarily required for coordination. Theoretical work by [Brav et al. \(2014\)](#) shows that investors can act in concert even without explicit interactions. However, such investors should still have correlated portfolios.

⁸ It is easy to check whether a group of institutions is in a clique. However, identifying cliques in a network is a problem whose solution requires computing time that increases in nondeterministic polynomial (NP) time as a function of the network size. For this reason, the clique problem is termed an NP complete problem.

the total fraction of the firm owned by all cliques, the fraction of the firm owned by the single clique with the largest stake in the firm, and the concentration of clique ownership. These measures are defined and discussed in Section 3.

It is important to note that a clique's ownership in a firm need not be made of 5% individual stakes. Common 5% stakes are used to identify connected pairs of institutions, but in a given firm, the members of a given clique may not have any common 5% positions at all. For example, in our sample, Carl Icahn only owned 0.5% of Apple stock when he campaigned for higher payout in 2013. His 0.5% stake is counted as part of his clique's total ownership (4.9%). In fact, we can exclude the $\geq 5\%$ positions from clique ownership after we have used them to identify the cliques. Our results are virtually unchanged under this specification (see online Internet Appendix IA-4).

Given the legal framework in the US, it is reasonable to ask whether we should expect investors to coordinate. While McCahery et al. (2016) find that institutions consider the legal implications when they interact, anecdotal evidence suggests that coordination is common.⁹ Coordination is legal for institutions that jointly own less than 5%, or those that own more than 5% and file joint 13D's. These joint filings cover situations in which investors "agree to act together for the purpose of acquiring, holding, voting or disposing of equity securities of an issuer."¹⁰ However, coordination can be legal even when groups own more than 5% and do not file jointly.¹¹ Furthermore, it is legal for institutions to communicate their voting or trading intentions (absent sharing inside information). Therefore, it is reasonable to examine if coordination can be detected in the network of institutional ownership.

3. Data and summary statistics

We obtain institutional ownership data from Thomson Reuters 13F database. To generate annual, calendar year-end holdings data, we first filter out cases in which the manager reports multiple positions in the same stock on the same report date and use only the holdings with the latest filing date. We split adjust reported holdings if the split occurs between the report date and filing date.¹² Following Griffin and Xu (2009), we carry holdings forward one quarter if an institution misses a single reporting period. Finally we retain the holdings reported closest to the end of each calendar year for each institution if the report date is no earlier than October (or July in the case of a reporting gap).

⁹ See, e.g., Foley and McLannahan (2016), Das (2015), and Benoit and Grind (2015).

¹⁰ 17 CFR 240.13d-5.

¹¹ See, e.g., the ruling in CSX v. Children's Investment Fund Management (UK), 654 F.3d 276 (2d Cir. 2011), "[t]he Rule does not encompass all 'concerted action' with an aim to change a target firm's policies." We thank Chester Spatt for pointing us to this ruling.

¹² Thomson Reuters adjusts the number of shares held for splits that occur after the report date. To recover the number of shares held at the report date, we undo this adjustment using the CRSP share adjustment factor.

We present summary statistics for institution-level observations in Table 1. The average institution in our sample has \$17.6 billion in assets under management and 1137 stocks in their portfolio. Seventy four percent of our sample institutions are investment companies, 12% are banks, and the remaining are split between insurance companies, pensions, endowments, and unidentified (miscellaneous).

We compute firm-level measures of clique ownership for each year using the membership lists in which each institution is either assigned to a specific clique or to no clique at all in that year. The extent to which a given firm's ownership is made up of institutional cliques is the following:

$$\text{Clique ownership}_{j,t} = \sum_i^N \lambda_{i,t} \mathbb{1}(\text{Clique institution}_{i,t}), \quad (1)$$

where $\lambda_{i,t}$ is institution i 's percent holdings in firm j at time t , and $\mathbb{1}(\text{Clique institution}_{i,t})$ is an indicator variable set to one if the institution belongs to a clique. Ownership by all cliques in the firm should proxy for coordination given that prior research shows that networks with high numbers of cliques better support coordination, overall. In addition to the aggregate ownership by cliques for each firm, we measure the concentration of clique ownership in a firm. *Clique Herfindahl* _{j,t} takes the total fraction owned by each clique present in firm j , squares it and then sums across all cliques. Last, we compute the total ownership in each firm of only the single clique with the largest ownership stake in the firm, *Clique own.-top 1* _{j,t} .

Our sample consists of clique membership lists that cover 59,648 institution-year observations (including institution years that are not assigned to a clique) from 1980–2013. After requiring data on institutional type, we have 51,699 observations. We aggregate clique and nonclique ownership to the firm level and merge with Compustat and CRSP. After requiring lagged book equity from Compustat and lagged returns from CRSP, and removing firms with institutional ownership over 100%, we have 218,351 firm-year observations. In some tests we use voting data from ISS for the period 2003–2013. This includes both voting outcomes at the firm level, and for mutual funds, specific ballot level voting results. We obtain executive compensation data and CEO age from ExecuComp.

Table 1, Panel B, presents summary statistics at the firm level. *Clique ownership* is 0.29 for the average firm. There is substantial variation; the inter-quartile range is 0.03–0.63. *Clique ownership* is also related to other ownership structure variables. *Clique ownership* is high when the level of institutional ownership is high, but also when the concentration is low. This is suggestive of a possible substitution effect between ownership concentration and ownership by cliques. We present evidence of this substitution in Fig. 2, where we plot the average institutional ownership concentration (defined as in Hartzell and Starks, 2003) each year. While total institutional ownership has gone up dramatically over this period, so too has the number of institutions. As a result, the average position size of the top institutional owners in each firm has dropped dramatically over time. We contrast this with the ownership by cliques: *Clique ownership* has risen substantially

Table 1

Summary statistics.

This table presents summary statistics on institution year (Panel A) and firm year (Panel B) observations from 1980–2013. Institution-level variables are constructed using calendar year-end holdings of each institution reported by Thomson Reuters. *Clique institution* is a dummy variable equal to one if an institution is in a clique and zero otherwise. Institutional types (Bank, Insurance company, Public pension, Endowment, and Miscellaneous) are indicator variables. All other variables are defined in Appendix 7.

<i>Panel A: Institution-level sample</i>					
	Mean	Median	Std. dev	10th	90th
Clique institution	0.36	0.00	0.48	0.00	1.00
Assets under management (2013 \$Mil.)	17567.62	1898.24	99720.38	394.98	26439.53
Number of positions	1136.76	461.00	2143.02	118.00	2572.00
Average holding size	0.01	0.00	0.02	0.00	0.01
Investment company	0.74	1.00	0.44	0.00	1.00
Bank	0.12	0.00	0.33	0.00	1.00
Insurance company	0.04	0.00	0.20	0.00	0.00
Corporate pensions	0.02	0.00	0.14	0.00	0.00
Public pensions	0.01	0.00	0.10	0.00	0.00
Endowments	0.01	0.00	0.09	0.00	0.00
Miscellaneous	0.06	0.00	0.23	0.00	0.00
Observations	53,178				
<i>Panel B: Firm-level sample</i>					
	Mean	Median	Std. dev	10th	90th
Clique ownership	0.29	0.21	0.29	0.01	0.69
Clique Herfindahl	0.06	0.01	0.89	0.00	0.12
Cliques own.—top 1.	0.13	0.09	0.17	0.01	0.30
IO concentration	0.72	0.76	0.25	0.36	1.00
Institutional ownership	0.33	0.25	0.32	0.01	0.78
Number of stocks	536.25	318.60	614.29	15.60	1427.77
Number of blockholders	1.24	1.00	1.53	0.00	3.00
Dedicated	0.04	0.01	0.08	0.00	0.13
Quasi-Indexer	0.20	0.14	0.21	0.01	0.49
Transient	0.08	0.04	0.11	0.00	0.22
Assets of owners (2013 \$Mil.)	30856.07	6811.88	50430.55	140.99	106501.89
Assets (2013 \$)	7252.78	319.07	66592.60	23.08	7196.38
Book leverage	0.17	0.11	0.22	0.00	0.43
Ln(Market to book)	0.48	0.48	1.16	−0.63	1.74
Observations	218,352				

over this same period. Regarding the concentration of clique ownership, *Clique Herfindahl* is 0.06 on average. The clique with the largest aggregate position in a firm (*Clique own.—top 1*) owns 13% of the firm on average. If members of this clique are, in fact, acting as one, this is a substantial blockholding, one that has so far been overlooked in the governance literature.

In Table 2 we present characteristics of institutions that belong to cliques. We regress *Clique institution* on lagged institutional characteristics associated with the institution's investment objectives and portfolio holdings. These characteristics include indicators for institution type: investment company, insurance company, bank, miscellaneous (where pension funds are the omitted type.) We also include indicators for trading classification following Bushee (2001): dedicated, transient, and quasi-indexer (the omitted type). Finally, we include variables related to the size and portfolio of the institution. These include assets under management (AUM), number of positions, number of block positions and average holding size.

We use a linear probability model with time effects and standard errors clustered at the institution level to test the relation between these characteristics and clique membership. The point estimates on AUM are negative and

insignificant, indicating that there is no relation between portfolio size and the tendency to belong to a clique. It is therefore not the case that large investors that tend to have common overlapping positions with many others (e.g., Blackrock or Vanguard) are more likely to belong to cliques. This may be because a clique requires all pairwise connections to exist between all others, and large investors may simply be connected to too many others for a clique to exist. Controlling for AUM, clique members tend to have more positions and own larger positions in firms, on average. Most institution types (e.g., banks, endowments) are more likely to belong to cliques relative to pensions (the omitted type), a potentially interesting result given that pensions are traditionally thought of as activist investors (Smith, 1996). Institutions that belong to cliques are most likely to be dedicated investors and least likely to be quasi-indexers (as classified by Bushee, 2001).¹³

¹³ In general, clique members tend to be more similar to other members of the clique across characteristics than they are to institutions outside the clique. This is particularly true for institutional type indicators, e.g., endowments work with other endowments, etc. These results are presented in detail in the online Internet Appendix Table IA-1.1.

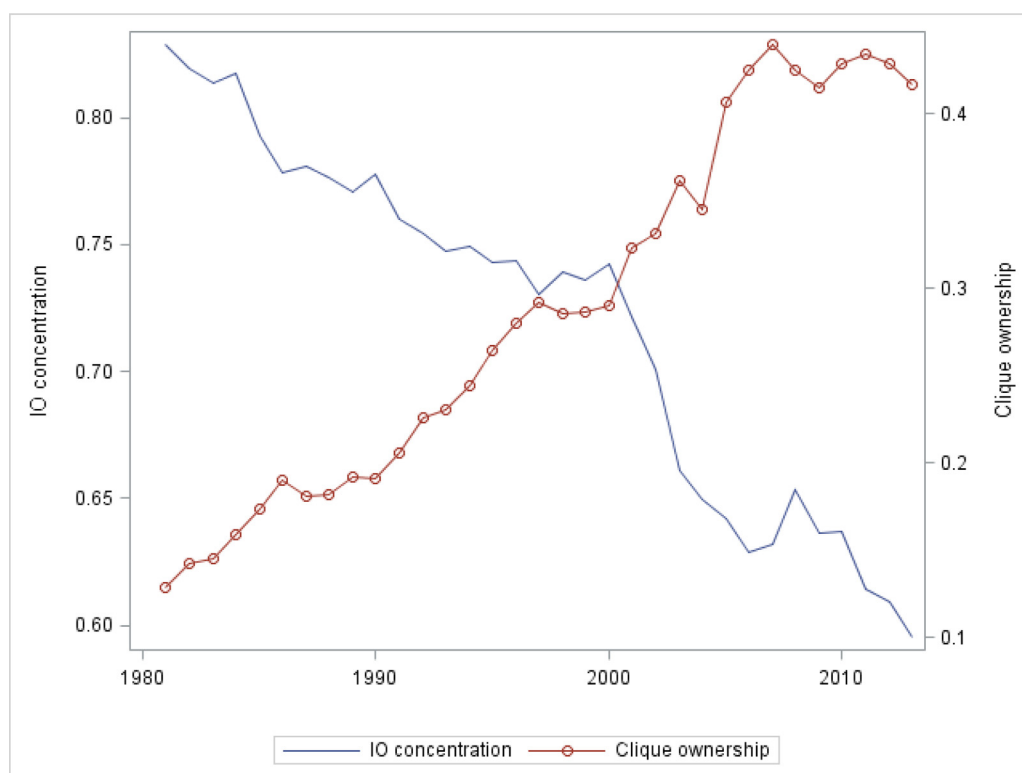


Fig. 2. Institutional ownership concentration and ownership by cliques over time. This figure presents the time series of cross-sectional means of the concentration of institutional ownership and the fraction of firms owned by cliques. *IO concentration* is the percentage owned by the five top institutions divided by institutional ownership as in [Hartzell and Starks \(2003\)](#), and *Ownership by cliques* is total ownership by cliques of institutional investors.

4. Cliques and governance via voice

Coordination should reduce the free rider problem. A reduction in free riding by owners should result in increased governance via voice (e.g., [Shleifer and Vishny, 1986](#)). If our measure of cliques reflects coordination by institutional investors, then we expect ownership by cliques to be positively associated with governance via voice. In this section, we test this hypothesis in several ways: through the examination of proxy voting, joint ownership filings, shareholder proposals, proxy fights, and corporate policies, such as cash payouts and divestitures.

4.1. Clique ownership and shareholder voting

Shareholder voting is a natural setting in which to examine the relation between shareholder coordination and governance. Voting is ultimately the way in which shareholders exercise their control rights. [McCahery et al. \(2016\)](#) find that institutions view voting as an engagement mechanism that directly affects governance.

First, we examine the extent to which owners vote together. If clique members coordinate, we expect two institutions' votes to have the same sign (for or against the proposal) more often when they are in the same clique than when they are not. This requires data that are only available for a subset of institutions—mutual funds. For each proxy ballot item, we create an indicator

variable equal to one if the mutual fund votes for, and zero otherwise. We then average this variable within the institution (the fund family) weighted by fund AUM.¹⁴ Using all unique pairs of institutions that vote on the same ballot item, we regress the voting variable of one randomly chosen institution from the pair on the other (the peer institution). We include an indicator equal to one if the two institutions are in the same clique (and zero otherwise), as well the interaction between this indicator and the peer vote. If either institution is not assigned to a clique, then the indicator takes a value of zero. These results are presented in [Table 3](#).

The significant positive coefficient on *Peer vote* represents the linear relationship between the votes of institutions not in the same clique. In general, mutual funds not in the same clique vote in the same direction between 44% and 51% of the time. If two funds are in the same clique, the correlation is increased by 8–10%. This is shown by the interaction term, *Same clique* $\times$ *Peer vote*, which is statistically significant at the one percent level. The negative coefficient on *Same clique* shows that institutions pairs within the same clique vote in favor of ballot items less frequently, all else equal. These findings are robust

¹⁴ In general, there is very little variation within mutual fund families in terms of votes, but for some family-item pairs, the average falls between zero and one when all funds managed by an institution do not vote the same.

Table 2

Characteristics of institutions in cliques.

This table presents a linear probability model of the probability that a given institution is in a clique. The sample is institution-year observations from 1981 to 2013 and is constructed using calendar year-end holdings of each institution reported by Thomson Reuters. All independent variables are lagged. AUM is the total market value of the institution's holdings in millions. Number of positions is the total number of stock holding in the portfolio. Number of large positions is the number of holdings with a stake of at least 5% in firm. Average percent of firm owned is the percent of the firm's market value owned by the institution averaged over all positions in the institution's portfolio. Dedicated and Transient are indicator variables defined as in [Bushee \(2001\)](#). Year effects are included but not reported. Standard errors are clustered by institution and reported in parenthesis with significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1) In a clique	(2) In a clique	(3) In a clique
AUM	−0.049 (0.11)	−0.073 (0.12)	−0.050 (0.12)
Number of positions	0.048*** (0.00)	0.049*** (0.00)	0.049*** (0.00)
Number of large positions	0.978* (0.51)	1.004* (0.52)	0.938* (0.50)
Average holding size	9.560*** (1.05)	10.452*** (0.97)	9.606*** (1.06)
Dedicated institutions	0.156*** (0.03)		0.157*** (0.03)
Transient institutions	0.088*** (0.01)		0.084*** (0.01)
Investment company		0.138*** (0.03)	0.118*** (0.03)
Insurance company		0.115*** (0.04)	0.106*** (0.04)
Bank		0.125*** (0.03)	0.123*** (0.03)
Endowments		0.046 (0.08)	0.027 (0.08)
Miscellaneous		0.110*** (0.04)	0.089** (0.04)
Constant	0.298*** (0.02)	0.205*** (0.04)	0.191*** (0.04)
Observations	51,699	51,699	51,699
Year effects	Yes	Yes	Yes

Table 3

Correlated voting within cliques.

Each mutual fund ballot item pair is set equal to one if the mutual fund votes in favor of the ballot item, and zero otherwise. For each ballot item, these indicators are aggregated to the institution level (weighted by funds' AUM). For each unique pair of institutions that vote on a particular ballot item, we regress the institution's vote on that of the other institution (*Peer vote*). *Same clique* is an indicator equal to one if the two institutions in the pair are in the same clique, and zero otherwise. Included fixed effects are indicated at the bottom of the table. Standard errors are clustered by institution and reported in parenthesis. Significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1) Vote	(2) Vote	(3) Vote	(4) Vote	(5) Vote
Same clique × Peer vote	0.038*** (0.01)	0.038*** (0.01)	0.036*** (0.01)	0.035*** (0.01)	0.043*** (0.00)
Peer vote	0.501*** (0.01)	0.499*** (0.01)	0.513*** (0.01)	0.511*** (0.01)	0.435*** (0.01)
Same clique	−0.033*** (0.01)	−0.031*** (0.01)	−0.033*** (0.01)	−0.030*** (0.01)	−0.034*** (0.00)
Year effects	No	Yes	No	Yes	No
Inst. effects	No	No	Yes	Yes	No
Meeting effects	No	No	No	No	Yes

to the inclusion of year effects and fixed effects for both institutions in the pair. They also hold using only variation within specific shareholder meetings as evidenced in column 5 where we include meeting fixed effects.

While we find that mutual fund families in the same clique vote together, this represents only a subsample of all institutional investors. This is potentially important given that cliques are composed of different types of institutions.

We can, however, observe the overall voting outcome for each ballot item at the firm level. These overall voting outcomes include votes from all types of institutions, not just mutual funds. Therefore, we use these data to determine if clique ownership impacts voting at the firm level.

[Gillan and Starks \(2000\)](#) show that activist governance is associated with more votes against management recommendations. [Matvos and Ostrovsky \(2010\)](#) show that

Table 4

Clique ownership and shareholder voting.

The dependent variable is the percentage of votes against management's recommendation. Panel A presents results using ISS recommendations as the measure for proposal quality. The measure of coordinated ownership is *Clique ownership* in columns 1 and 2, *Clique Herfindahl* in columns 3 and 4, and *Clique own.—top 1* in columns 5 and 6. The variables are defined in Appendix 7 Columns 1, 3, and 5 use the sample of director election ballot items proposed by management. The remaining columns use all other ballot items proposed by management. Panel B presents results using proposal quality as in Davis and Kim (2007). Ownership variables are measured as of December of the year prior to the year of the shareholder meeting. Market to book is measured at the most recent fiscal year end prior to the meeting. Stock returns and firm size are measured at the month prior to the meeting. Numbers of blockholders, institutional owners, and stocks in the owners portfolio are reported per 1000 for ease of interpretation. All regressions include year and firm effects. Standard errors are clustered by firm with standard errors reported in parenthesis and significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Panel A: Proposal quality based on ISS recommendations						
	(1) Per. votes against	(2) Per. votes against	(3) Per. votes against	(4) Per. votes against	(5) Per. votes against	(6) Per. votes against
Clique ownership _{<i>t</i>-1}	-0.072*** (0.02)	-0.014 (0.02)				
Clique ownership _{<i>t</i>-1} × Bad proposal _{ISS}	0.213*** (0.01)	0.171*** (0.01)				
Clique Herfindahl _{<i>t</i>-1}			-0.062*** (0.02)	-0.069*** (0.01)		
Clique Herfindahl _{<i>t</i>-1} × Bad proposal _{ISS}			0.273*** (0.06)	0.201*** (0.04)		
Cliques own.—top 1 _{<i>t</i>-1}					-0.029*** (0.01)	-0.036*** (0.01)
Cliques own.—top 1 _{<i>t</i>-1} × Bad proposal _{ISS}					0.244*** (0.03)	0.111*** (0.02)
Bad proposal _{ISS}	0.016*** (0.00)	0.069*** (0.01)	0.103*** (0.00)	0.143*** (0.00)	0.075*** (0.01)	0.137*** (0.01)
Institutional ownership _{<i>t</i>-1}	0.073*** (0.02)	0.041** (0.02)	0.042*** (0.01)	0.057*** (0.01)	0.036*** (0.01)	0.055*** (0.01)
Dedicated _{<i>t</i>-1}	0.012 (0.01)	0.001 (0.01)	0.009 (0.01)	0.001 (0.01)	0.010 (0.01)	-0.000 (0.01)
Transient _{<i>t</i>-1}	-0.025*** (0.01)	-0.002 (0.01)	-0.022*** (0.01)	-0.008 (0.01)	-0.019** (0.01)	-0.007 (0.01)
Num. of stocks in owners' portfolio _{<i>t</i>-1}	0.011*** (0.00)	0.004 (0.00)	0.012*** (0.00)	0.003 (0.00)	0.012*** (0.00)	0.003 (0.00)
Number of inst. owners _{<i>t</i>-1}	-0.008 (0.01)	0.028** (0.01)	-0.008 (0.01)	0.025* (0.01)	-0.006 (0.01)	0.026* (0.01)
Own. of top 5 _{<i>t</i>-1}	-0.050*** (0.01)	-0.036*** (0.01)	-0.040*** (0.01)	-0.018 (0.01)	-0.050*** (0.01)	-0.020 (0.01)
Num. of blockholder _{<i>t</i>-1}	0.001*** (0.00)	-0.000 (0.00)	0.001** (0.00)	-0.000 (0.00)	0.001** (0.00)	-0.000 (0.00)
Market to book (bps) _{<i>t</i>-1}	-0.001 (0.00)	0.002*** (0.00)	-0.001 (0.00)	0.002*** (0.00)	-0.002 (0.00)	0.002*** (0.00)
Ln(Size) _{<i>t</i>-1}	-0.004*** (0.00)	-0.007*** (0.00)	-0.004*** (0.00)	-0.007*** (0.00)	-0.004*** (0.00)	-0.007*** (0.00)
Assets of owners (\$Tril.) _{<i>t</i>-1}	-0.035 (0.02)	0.008 (0.03)	-0.055** (0.02)	-0.002 (0.03)	-0.047* (0.02)	-0.002 (0.03)
Observations	128,091	45,142	128,091	45,142	128,060	45,129
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes	Yes	Yes	Yes
Meeting type	All	All	All	All	All	All
Vote type	Director	Nondirector	Director	Nondirector	Director	Nondirector

Panel B: Proposal quality based on Davis and Kim (2007) classification

	(1) Per. votes against	(2) Per. votes against	(3) Per. votes against
Clique ownership _{<i>t</i>-1}	0.010 (0.02)		
Clique ownership _{<i>t</i>-1} × Good proposal _{DK2007}	-0.030*** (0.00)		
Clique Herfindahl _{<i>t</i>-1}		-0.048*** (0.01)	
Clique Herfindahl _{<i>t</i>-1} × Good proposal _{DK2007}		-0.130*** (0.02)	

(continued on next page)

Table 4 (continued)

Panel B: Proposal quality based on Davis and Kim (2007) classification			
	(1) Per. votes against	(2) Per. votes against	(3) Per. votes against
Cliques own.—top 1 _{t-1}			−0.024*** (0.01)
Cliques own.—top 1 _{t-1} × Good proposal _{DK2007}			−0.059*** (0.01)
Good proposal _{DK2007}	−0.161*** (0.00)	−0.161*** (0.00)	−0.161*** (0.00)
Institutional ownership _{t-1}	0.040** (0.02)	0.059*** (0.01)	0.057*** (0.01)
Dedicated _{t-1}	0.003 (0.01)	0.001 (0.01)	0.001 (0.01)
Transient _{t-1}	−0.003 (0.01)	−0.009 (0.01)	−0.008 (0.01)
Num. of stocks in owners' portfolio _{t-1}	0.005 (0.00)	0.003 (0.00)	0.004 (0.00)
Number of inst. owners _{t-1}	0.029** (0.01)	0.025* (0.01)	0.027** (0.01)
Own. of top 5 _{t-1}	−0.030** (0.01)	−0.016 (0.01)	−0.017 (0.01)
Num. of blockholder _{t-1}	−0.001 (0.00)	−0.000 (0.00)	−0.000 (0.00)
Market to book (bps) _{t-1}	0.002*** (0.00)	0.002*** (0.00)	0.002*** (0.00)
Ln(Size) _{t-1}	−0.007*** (0.00)	−0.007*** (0.00)	−0.007*** (0.00)
Assets of owners (\$Tril.) _{t-1}	−0.009 (0.03)	−0.008 (0.03)	−0.010 (0.03)
Observations	45,142	45,142	45,129
Year effects	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes
Meeting type	All	All	All
Vote type	Nondirector	Nondirector	Nondirector

mutual funds are more likely to vote against management when other funds are more likely to vote against management. We complement and extend these analyses. If cliques improve governance via voice, we expect voting outcomes to differ at the firm level as a function of clique ownership. This does not necessarily mean they should vote against proposals, on average. Informed and coordinated investors should vote in favor of high quality proposals and against those that destroy value.

To test this prediction, we proxy for the quality of proposals in two ways. To proxy for low quality proposals, we use proposals that ISS has recommended against, *Bad proposal*_{ISS}. ISS recommendations are a standard proxy for quality in the literature (e.g., [Bethel and Gillan, 2002](#); [Cai et al., 2009](#); [Cotter et al., 2010](#); [Morgan et al., 2006](#); [Morgan et al., 2011](#)). We proxy for good proposals, *Good proposal*_{DK2007}, using six items identified in [Davis and Kim \(2007\)](#) (and used in [Morgan et al., 2011](#)) as having the most significant positive impact on shareholder value. These include proposals for (1) declassifying the board, (2) establishing cumulative voting, (3) establishing an independent chairman of the board, (4) repealing shareholder rights plans (poison pills), (5) giving shareholders voice on golden parachutes, and (6) expensing stock options.

We present our tests of the relation between clique ownership and voting in [Table 4](#). Our independent variable is the percentage of votes against management, and our

variables of interest are our measures of clique ownership and their interactions with proposal quality. Panel A presents results using the ISS recommendations as our proposal quality proxy, and Panel B presents results using the proxy based on [Davis and Kim \(2007\)](#). All regressions include firm and year fixed effects as well as a variety of controls for other ownership and firm characteristics. This ensures that the effects we measure related to clique ownership are different from total institutional ownership, blockholder ownership, index-type institution ownership, etc. All explanatory variables are measured the year before the election event.

We find that clique ownership is positively related to voting against bad management proposals. When ISS disagrees with management, i.e., proposal quality is likely to be poor, coordinated ownership is associated with an increase in the number of votes against, suggesting that coordinated ownership goes against the passage of these proposals. This is true for director elections (columns 1, 3, and 5) and other ballot items (columns 2, 4, and 6). The direction of this effect is consistent across all three proxies for coordinated ownership (ownership level of the cliques, the Herfindahl of clique ownership, and the ownership of the top clique). The effect is statistically significant at the 1% level in all six specifications.

The economic magnitudes of these effects are large. From column 1, firms with *Clique ownership* one standard

deviation above the mean have 6.2 percentage points more votes against management when ISS recommends against. This is economically large compared to the average percentage of votes against management of 5.8%. The concentration of clique ownership is also strongly related to votes against bad management proposals. A firm with *Clique Herfindahl* one standard deviation above the mean is associated with 24 percentage points more votes against management in director elections when ISS also recommends against (column three), roughly five times the unconditional average. The economic effect of ownership by the clique with the largest position in the firm (column 5) is similar to that of *Clique ownership*. It is worth pointing out that, for these results to be driven by unobserved firm or ownership characteristics, such factors would have to be time varying and unrelated to standard measures of ownership structure, such as the level of institutional ownership, the number of blockholders, the ownership of the top five institutions, etc.¹⁵

We also find that clique ownership increases support for good management proposals (Table 4, Panel B). The point estimate of the interaction between our measures of clique ownership and *Good proposal*_{DK2007} are all negative and statistically significant at the 1% level.¹⁶ This means that when management proposals are good, high clique ownership is associated with more shareholder support. The effect is consistent with what we observe in Panel A, but the economic magnitude is smaller. This is not surprising. Unconditionally, most votes go in favor of management, therefore coordinated ownership has a smaller impact on increasing votes in favor. A one standard deviation in *Clique ownership* associated with slightly less than a one percentage point decrease in votes against. The other measures of clique ownership have slightly larger estimates. A one standard deviation increase in *Clique Herfindahl* and the ownership of the top clique are associated with more than a 10% decrease in votes against, relative to the unconditional mean. Our findings are consistent with an interpretation that ownership by cliques is associated with stronger governance through shareholder voting.

4.1.1. Exogenous changes to the network and voting

Our findings on the effects of clique ownership on voting are subject to alternative interpretations. Clique ownership can form endogenously in expectation of future improvements in governance, or clique ownership and governance can be jointly determined by a firm's unobservable time-varying characteristics. For example, clique members can self-select into firms with ownership that systematically opposes bad management proposals and supports good management proposals more often. To alleviate these concerns, we use the 2003 mutual fund

scandal as a shock to clique ownership. In September 2003, 25 mutual fund families were accused of illegal trading practices by the SEC and New York State Attorney General Eliot Spitzer. This scandal resulted in significant outflows from and closures of institutions that impacted connections within the network of institutional investors and reshaped existing cliques. These changes in ownership allocations took place from 2003 through 2005.

To instrument for ownership by cliques, we first identify the investors who were not themselves implicated in the scandal, but were connected to scandal institutions. For each institution i , we use the holdings data from the third quarter of 2003 to measure the fraction of the institution's connections that are with scandal funds, where a connection is deemed to exist if the two institutions have overlapping 5% holdings. We then aggregate this institution-level treatment identifier to the firm level such that treatment firms are those owned to a high degree by institutions that are highly connected to institutions implicated in the scandal. Specifically, we define firm-level treatment for firm j as:

$$Treatment_j = \sum_i \lambda_{i,j} \left(\frac{1}{|C_i|} \sum_{k \in C_i} \mathbb{1}_k \right), \quad (2)$$

where $\lambda_{i,j}$ is institution i 's percent holding in firm j , N is the total number of institutions, $C_i \subseteq N$ is the set of all institutions connected to institution i , $|C_i|$ is the cardinality of C_i (i.e., number of institutions connected to institution i), and $\mathbb{1}_k$ is an indicator equal to one if institution k is a scandal institution. Using this measure as a continuous treatment identifier, we instrument for clique ownership using changes in the institutional ownership network resulting from the scandal.

Our instrument satisfies the exclusion restriction if ownership by scandal mutual funds only affects voting on management proposals through their effect on firm-level clique ownership. Based on the results in Zitzewitz (2009) that suggest that prosecutions were, to a degree, a result of the headquarters location of the institution, such an assumption seems plausible. The same shock has been used to provide plausibly exogenous variation to common ownership by Anton and Polk (2014) and Koch et al. (2016). It is unlikely that changes to the network induced by the scandal are the result of future voting outcomes in specific firms. To ensure that there are no direct effects of large holdings by nonclique scandal mutual fund ownership, we control for this type of ownership in all regressions, including first stage regressions for our two instrumented variables.

We find that institutions with stronger exposure to the scandal institutions experienced a decrease in the likelihood of being in a clique during 2003–2005. As a result of this decrease in the exposed institutions' probability of being a clique, firms that these institutions owned experienced an exogenous decrease in *Clique ownership*. We exploit this variation and show a difference in voting against management after the scandal (*Post*) for firms with exogenously lower clique ownership (*Treatment*). First, we estimate first stage regressions for both the relation between clique ownership and the exogenous treatment

¹⁵ Our dependent variable, the fraction of votes against management, is bounded between zero and one, so we can face a censoring problem of some latent voting outcome. Our conclusions are unchanged when we estimate the effect using a Tobit model. However, due to the nonlinear nature of the Tobit model, under this specification we cannot include firm fixed effects.

¹⁶ We do not present results from director elections separately, as our proxy for good proposals, *Good proposal*_{DK2007}, applies only to other agenda items.

Table 5

Clique ownership and shareholder voting: exogenous network shocks.

The dependent variable is the percentage of votes against management's recommendation. Treatment firms are those owned to a high degree by institutions whose network was affected by the mutual fund late trading scandal in 2003. The top row presents the estimate of the main effect from the first stage where the instrument is an indicator for scandal exposed firms (*Treatment*) interacted with the period after the scandal (*Post*). We also instrument for the interaction of clique ownership and ISS using the interaction of *Treatment* × *Post* with ISS. First stage estimates of the interaction term are suppressed here for space but shown in the online Internet Appendix. Results from the second stage are presented below. Column 1 uses the sample of director election ballot items proposed by management. Column 2 uses all other ballot items proposed by management. Column 3 uses a measure of proposal quality from Davis and Kim (2007) and presents results for the nondirector election sample. Numbers of blockholders, institutional owners, and stocks in the owners portfolio are reported per 1000 for ease of interpretation. Control variables as in Table 4 are included but suppressed for space. All regressions include year and firm effects. Standard errors are clustered by firm with standard errors reported in parenthesis and significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1) Clique own.	(2) Clique own.	(3) Clique own.
<i>First stage: main effect</i>			
Treatment × Post	−1.918*** (0.41)	−2.016*** (0.47)	−2.016*** (0.47)
	Votes against	Votes against	Votes against
<i>Second stage</i>			
Clique ownership _{<i>t</i>−1}	0.728*** (0.28)	1.681*** (0.61)	4.273** (1.92)
Clique ownership _{<i>t</i>−1} × $\widehat{\text{Bad proposal}}_{\text{ISS}}$	0.082*** (0.01)	0.238*** (0.05)	
Clique Ownership _{<i>t</i>−1} × $\widehat{\text{Good proposal}}_{\text{DK2007}}$			−4.641 (2.95)
Bad proposal _{ISS}	0.010** (0.00)	0.090*** (0.02)	
Good proposal _{DK2007}			0.181*** (0.01)
Scandal fund IO	−0.110* (0.06)	−0.270* (0.16)	−0.972* (0.54)
Observations	19,507	4582	4582
First stage F-stat	7.126	7.114	1.857
Controls	Yes	Yes	Yes
Year effects	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes
Meeting type	All	All	All
Vote type	Director	Nondirector	Nondirector

as well as the interaction between clique ownership and proposal quality and exogenous treatment. Specifically, we estimate:

$$\text{Clique own.}_{j,t,l} = \phi_1(\text{Treat}_j \times \text{Post}_t) + \phi_2 \text{Treat}_j + X_{j,t} \psi + \delta_j + \theta_t + \mu_{j,t,l}, \quad (3a)$$

$$\begin{aligned} \text{Clique own.}_{j,t,l} \times \text{Prop. qual.}_{j,t,l} &= \xi_1(\text{Treat}_j \times \text{Post}_t \times \text{Prop. qual.}_{j,t,l}) \\ &+ \xi_2 \text{Treat}_j + X_{j,t} \lambda + \delta_j + \theta_t + v_{j,t,l}, \end{aligned} \quad (3b)$$

where j indexes firms, t indexes years, and l indexes proposals. We include a vector of controls, X , as well as time (θ_t) and firm (δ_j) fixed effects. *Prop. qual.* is determined by either ISS recommendations or based on the content of the proposal as in Davis and Kim (2007), as described in Section 4.1.

We use these estimates to test whether clique ownership has an effect on voting outcomes. We estimate the second stage as follows:

$$\begin{aligned} \text{Votes against}_{j,t,l} &= \beta_1 \widehat{\text{Clique own.}}_{j,t,l} + \beta_2 (\widehat{\text{Clique own.}}_{j,t,l} \times \widehat{\text{Prop. qual.}}_{j,t,l}) \\ &+ \beta_3 \text{Treat}_j + X_{j,t} \gamma + \delta_j + \theta_t + \varepsilon_{j,t,l}. \end{aligned} \quad (4)$$

We present results from these specifications in Table 5. Columns 1 and 2 correspond to columns 1 and 2 of Table 4 Panel A. Column 3 corresponds to column 1 of Table 4 Panel B. The top panel of this table presents the first stage estimate on the main effect. Full first stage specifications are presented in online Internet Appendix Table IA-1.2. The first stage estimates for the main effect are economically and statistically significant at the 1% level. A one standard deviation increase in our treatment variable is associated with a 3% decrease in clique ownership relative to the prescandal level. Similarly, the interaction, *Clique ownership* × *ISS against mgmt* is 7% lower for a one standard deviation increase in the instrumented interaction. This interaction instrument is statistically strong for the director elections (1% level), but weaker for the smaller nondirector samples.

Consistent with prior results, the interaction between *Clique ownership* and *ISS recommending against management* is positively related to voting against management. This interaction effect represents an increase in votes against management over and above the effect of clique ownership when proposals are good. The magnitude is a 25% increase over the mean votes against, and is statistically significant at the 1% level (column 1). In total, we

estimate votes against management to be 18.5 percentage points higher when clique ownership is high and proposals are bad, compared to cases when clique ownership is low and proposals are good.¹⁷ For nondirector elections (column 2), this effect is even stronger. A one standard deviation increase in *Clique ownership* when ISS recommends against causes an increase in votes against management of four percentage points, which is also significant at the 1% level. In column 3, we show that using the instrumental variable (IV) framework, firms with high clique ownership vote against good proposals less often (27% fewer votes against). This is an economically large effect that is significant at the 11% level. One difference in these results from Table 4 is that, when using exogenous variation, *Clique ownership* is unconditionally positively related to voting against management, consistent with Gillan and Starks (2000). Overall, these results support the view that ownership by cliques causes more voting against management, particularly when proposals are of low quality.

There is evidence in the literature that institutions can vote in ways that do not maximize shareholder value, due to private benefits from conflicts of interest. While Del Guercio and Hawkins (1999) and Davis and Kim (2007) suggest conflicts of interest are not a problem generally, Butler and Gurun (2012) show that, when fund managers and CEOs share educational ties, voting outcomes may not maximize shareholder value. This can result in arguably worse governance outcomes due to a quid pro quo in CEO pay. In general, however, a conflict of interest alternative, such as that in Butler and Gurun (2012), would result in more votes in favor of management, which we do not find.

Furthermore, institutions might vote against other shareholders' interests if investors' horizons differ. An institution might vote myopically if it owns a small stake, and therefore it does not have the incentive to collect information on firm intangibles and to value long-term projects (Bushee, 1998). We do not believe that myopic voting is driving our results. First, ISS recommendations would need to also be myopic. This is unlikely because the organization itself serves to solve problems of dispersed ownership. Second, cliques tend to own large stakes, and if clique members operate as a single large block, they should have an incentive to value the long-term firm intangibles. On balance, we believe our results are consistent with improvements in governance.

Last, it is possible, and perhaps likely, that managers adjust proposal quality in response to exogenous changes in ownership structure, as the threat of intervention has changed. As long as the managerial response is imperfect, we can still identify relations between clique ownership and voting against management. Given the difficulty of identifying firm-level coordination outcomes from a shock that propagates throughout the entire network, we think it is unlikely that managers will perfectly adjust.

4.2. Clique ownership and joint ownership filings

In certain cases, institutions are required to disclose activist intentions if they own 5% or more of the firm. When engaging in activism in concert with others, owners are often required to make joint disclosures. These come in the form of SEC filings on Schedule 13D. This is arguably direct evidence of coordinated activism. In this section, we test whether clique ownership is related to disclosure of joint activism. Such tests are, however, limited in that joint filings alone will underestimate coordination efforts. Not all governance-related actions require joint filings, even if a group is working together and owns more than 5%.¹⁸ Recent legal rulings have pointed out that even some interactions beyond voting communications do not require joint disclosures of interest.¹⁹ This is supported by a recent number of anecdotes of investors coordinating, even in the absence of joint ownership disclosures (see, for example, recent news articles by Foley and McLannahan, 2016; Das, 2015; Benoit and Grind, 2015).

Nonetheless, we expect joint activism disclosures to be more prevalent for members of cliques. To test this, we take all institutions in our sample in 2003 and manually collect all of their 13D filings for 2003 and 2005. For each filing, we determine whether the institution is acting individually in the given company, or is filing jointly with some other investor(s). We then examine if the propensity to file jointly is related to clique membership. We present these results in Table 6. In this table, we regress the log number of joint filings on an indicator for whether the filing institution is in a clique. We control for the number of solo filings as well as a variety of institution level characteristics. We find strong statistical evidence that clique membership is positively associated with the number of joint filings. The effect is large and statistically significant at the 1% level. Being in a clique is associated with a 10% increase in the number of joint filings. It increases the probability of filing any joint 13D by 40%. This evidence supports the view that clique members are more likely to engage in explicit, disclosed, coordinated actions. It is important to note that institutions are not randomly assigned to cliques; therefore, it is possible that these associations could be driven not by clique membership but by omitted institutional characteristics or by selection of joint 13D filers into cliques. Unfortunately, we cannot use the identification strategy in Section 4.1.1. While the shock provides substantial variation in firm-level ownership structure, only about 15% of institutions enter or leave cliques between 2003 and 2005. Given that 13Ds are rare in the data, we have little power to test institution-level outcomes.

¹⁷ A one standard deviation change in clique ownership is 0.228 in the sample used in this regression. Therefore, the economic effect is $(0.728 + 0.082) \times 0.228 = 0.185$.

¹⁸ For example, after 1992, institutions are specifically allowed to communicate during a proxy contest about their voting intentions without the need to file jointly.

¹⁹ See, for example, the ruling in *CSX v. Children's Investment Fund Management*, (UK), 654 F.3d 276 (2d Cir. 2011), "[t]he Rule does not encompass all 'concerted action' with an aim to change a target firm's policies."

Table 6

Joint filings.

The sample is a sample of 13Ds in 2003 and 2005 for all institutions in the data in 2003. The dependent variable is equal the log number of joint 13D filings for an institution. *Clique institution* is equal to one if the institution is in a clique, and zero otherwise. $\ln(1+\text{Solo } 13D)$ is the number of solo 13D filings by an institution. *AUM* is assets under management in billions. $\ln(1+\text{Num. stocks})$ is the number of stocks in the institution's portfolio. Significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	Dependent variable: $\ln(1+\text{Num. joint13D})$			
	(1)	(2)	(3)	(4)
In a clique	0.109*** (0.01)	0.095*** (0.01)	0.082*** (0.01)	0.101*** (0.01)
$\ln(1+ \text{Solo } 13D)$		0.117*** (0.03)	0.088** (0.04)	0.108*** (0.03)
AUM			−0.000 (0.00)	0.000 (0.00)
$\ln(1+\text{Num. stocks})$			0.001 (0.00)	−0.008** (0.00)
Quasi-indexer			−0.014* (0.01)	
Dedicated			0.212*** (0.07)	
Corp. pension				0.047 (0.04)
Investment co.				0.026* (0.01)
Insurance co.				0.110*** (0.04)
Bank				−0.002 (0.01)
Public pension				−0.004 (0.02)
Misc.				0.020 (0.02)
Constant	0.017*** (0.00)	0.015*** (0.00)	0.019 (0.02)	0.044* (0.02)
Observations	4,150	4,150	3,828	3,828
R ²	0.030	0.044	0.063	0.049

4.3. Clique ownership and shareholder proposals and proxy fights

We examine if clique ownership relates to the frequency of shareholder proposals and proxy fights against management proposals. We obtain data from ISS on whether or not a ballot item was proposed by shareholders or management to identify the frequency of shareholder proposals. To measure proxy fights, we collect data on the frequency of filing SEC forms PREC14A, PREN14A, PRRN14A, DEFC14A, DEFN14A, DFRN14A, DFAN14A, and DEFC14C (following Norli et al., 2015).

We estimate the probability of either a shareholder proposal or a proposal proxy fight and present the results in online Internet Appendix Table IA-1.3. Using a conditional logit specification, we find the probability of a shareholder proposal is increasing in two of three measures of coordinated ownership. Specifically, a one standard deviation increase in *Clique Herfindahl* is associated with a probability of a shareholder proposal that is 2.5 times larger. Similarly, a one standard deviation increase in *Clique own.-top 1* is associated with a probability of a shareholder proposal that is 1.3 times larger. These results are significant at the 10% and 5% levels, respectively.

Next we examine proxy fights. Norli et al. (2015) point out that restrictions on ballot access are significant, and

therefore they propose using the SEC filings described above as a test of activism. Unfortunately, we identify only 111 activism events for our sample firms over 2004–2013 (compared to 385 from 1993–2007 in Norli et al., 2015). As a result, we have little power to detect relations between activism and coordination. Nevertheless, we present estimates of the probability of activism as a function of clique ownership in online Internet Appendix Table IA-1.3, columns 4–6. While the point estimates are consistently positive, given the few number of events, it is unsurprising the results are statistically insignificant. We do not have enough proxy fights and ballot proposals over the two years surrounding the network shock described in Section 4.1.1 to identify causal effects in this section. Therefore these results represent associations.

4.4. Clique ownership and corporate outcomes

Anecdotal evidence suggests that groups of investors coordinate to achieve specific corporate actions. This can be because they share similar views on governance, or because some institutions can be subject to prudent investor rules and have preferences for certain policies, like payout. We select several firm outcome variables that are associated with activist governance. We examine the extent to which each institution owns firms that

(i) initiate a dividend, (ii) initiate a repurchase, (iii) divest either through a spin off or by selling the entire firm, and (iv) make acquisitions. We identify dividend and repurchase activity from CRSP and Compustat and divestitures and acquisitions from SDC.

We create indicator variables for each of these four outcomes, and we aggregate these for each institution based on its holdings. For example, we define $d_{i,t}$ as an indicator set to one if firm i initiates a dividend in year t , and represent an institution's tendency to own firms that initiate dividends in period t by $\sum \lambda_{i,t} d_{i,t}$, where $\lambda_{i,t}$ is the institution's portfolio weight in stock i at time t . We do the same for the other three outcome variables.

After generating these institution-year measures, we then compute medians across all institutions within each clique in each year, resulting in measures of the extent to which each individual clique tends to be associated with these firm outcomes. We plot these yearly clique characteristics in Fig. 3. Each solid dot represents the median characteristic of a unique clique. The shaded diamond indicates the median for all institutions that do not belong to a clique. For example, in Panel (A) of Fig. 3, the shaded diamond in the year 2000 indicates that among institutions that do not belong to a clique, the median institution has a little over 1% of its portfolio in firms that initiated a dividend during that year. The solid dots indicate that all but one of the individual cliques have a larger fraction of their ownership in firms that initiated dividends.

There are two observations worth noting from these figures. First, there is considerable variation across cliques in their tendencies to own firms that exhibit these characteristics. Second, it is clear that for both dividends and acquisitions, clique ownership differs systematically from nonclique ownership. Panel (A) shows that the firms owned by cliques are much more likely to initiate dividends compared to firms owned by institutions that are not members of cliques. Similarly, in Panel (D), it is clear that most cliques own firms that do not make acquisitions compared to firms owned by institutions that do not belong to cliques. These findings are consistent with cliques either improving agency problems through governance or choosing firms that, in the future, exhibit behavior consistent with a reduction in agency problems.

Next we test if the cliques that tend to be strongly associated with, for example, initiating firm payouts in one period, are the same cliques associated with initiating firm payouts in subsequent periods. We first define a specialized clique as any clique in the top (or bottom, in the case of making acquisitions) quartile along a given corporate outcome in a given year. For instance, cliques that specialize in initiating dividends are represented by the solid dots that are in the top quartile in a given year in Panel (A) of Fig. 3. Similarly, cliques that specialize in preventing empire-building acquisitions are represented by the solid dots in the bottom quartile in a given year in Panel (D).²⁰

We expect ownership by cliques to have predictive power for future firm outcomes associated with activist governance. To test this, we measure the total fraction of each firm owned by specialized cliques separately for dividends (Dividend clique ownership), repurchases (Repurchase clique ownership), acquisitions (Anti-acquisition clique ownership), and divestitures (Divestiture clique). We then examine future policy outcomes as functions of lagged ownership by specialized cliques.

Table 7 presents estimates of the effect of ownership of specialized cliques on future dividend initiations, repurchase initiations, acquisitions, and divestitures respectively. In these regressions, we continue to include overall ownership by coordinating investors (*Clique ownership*). Generally, we find that ownership by the specific type of clique is associated with changes in the firm policy along that dimension in the next period. For example, the first column estimates the effect of dividend clique ownership at t on dividend initiations at $t + 1$. We see that the probability of initiating a dividend is increasing in the ownership of the dividend cliques in the year before. The overall effect is economically small. A one standard deviation increase in Dividend clique ownership is associated with a less than 1% absolute increase in the probability of dividend initiation. However, this represents an increase of 4% over the unconditional probability of initiation. This effect is significant at the 10% level.

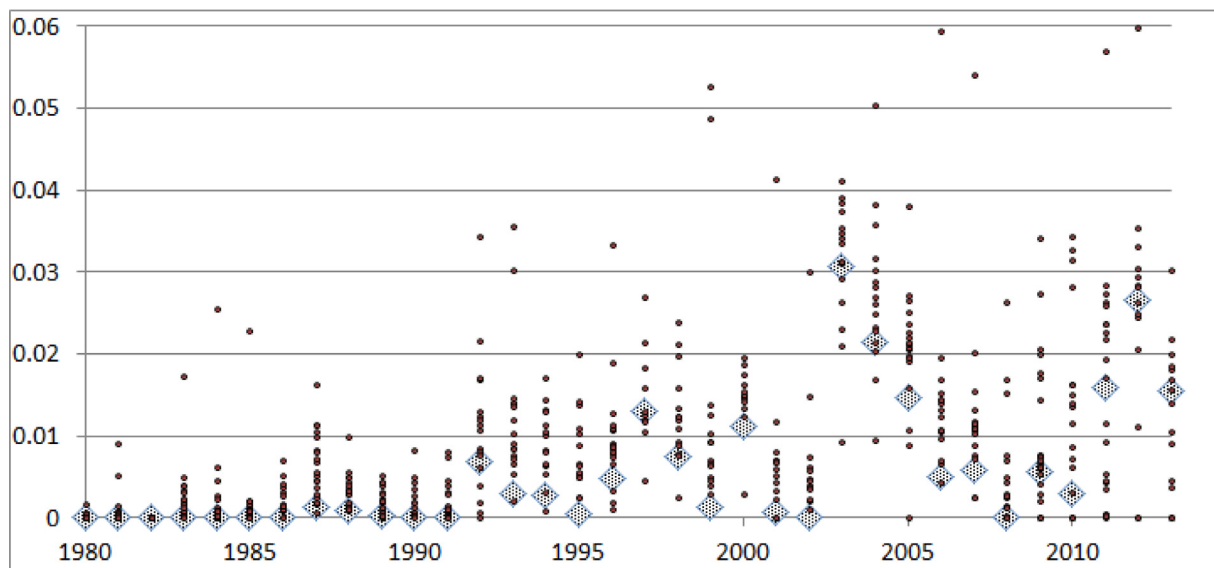
We see little evidence that clique ownership by institutions is related to repurchases. However, we find that ownership by anti-acquisition cliques is related to a lower probability of future acquisitions, and that ownership by cliques that specialize in divestitures is related to a higher probability of divestitures going forward, although these effects are also economically small. A one standard deviation increase in Anti-acquisition clique ownership is associated with a 2% decrease in acquisitions in the next year, relative to the unconditional mean probability of an acquisition. This estimate is significant at the 10% level. Results in column 4 indicate that dividend and anti-acquisition cliques also avoid firms that conduct divestitures in the future. This result is of similar magnitude and is also significant at the 10% level. Ownership by specialized cliques is not randomly assigned, and our network shock used in Section 4.1.1 does not provide sufficient variation in ownership by specialized cliques to identify causal effects here. Therefore, we cannot distinguish whether these results are picking up the fact that specialized cliques are better at forecasting future policy changes, or whether they are actually causing those changes. Both interpretations are of interest, as each suggests that specialized clique ownership predicts future financial policies.

5. Cliques and governance via exit

Coordination facilitates governance via voice (e.g., voting) by mitigating free riding incentives. However, Edmans and Manso (2011) show that the free rider problem actually helps governance through the threat of exit, and that coordination, by alleviating free riding, weakens this threat. This is because the threat of exit is strongest when owners trade independently and aggressively. If

²⁰ We find that membership in these specialized cliques is persistent. The likelihood that a firm remains in a specialized clique from one year to the next is roughly twice what would be expected due to chance and is statistically significant at the 1% level.

(A) Initiate dividends



(B) Initiate repurchases

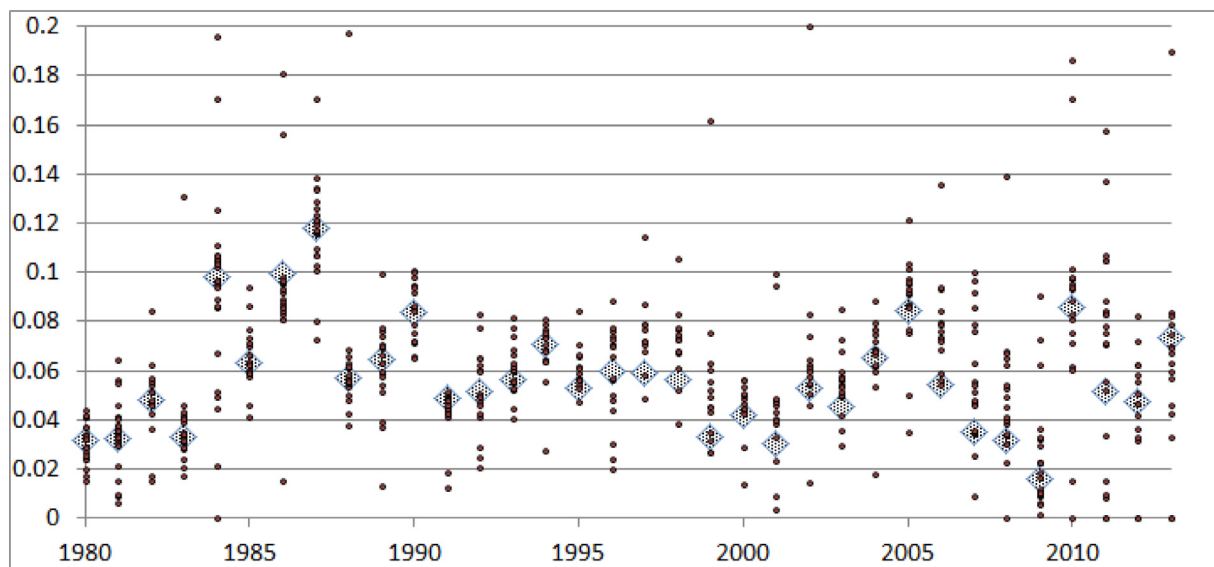


Fig. 3. Cliques and revealed preferences. For each clique and by year, we sort institutions by the fraction of their portfolios invested in firms with a specific corporate policy outcome (e.g., initiate a dividend). We then represent the clique's preference for this policy by the median value. Each clique is represented by a solid dot. The median value among institutions that do not belong to a clique is represented by the shaded diamond. In Panel (A), the y-axis indicates the fraction of the median institution's portfolio that is in firms that initiated a dividend that year. Panel (B) plots institutions' tendencies to own firms that initiate repurchases. Panel (C) plots the tendency to own firms that initiate divestitures, and Panel (D) plots firm acquisitions.

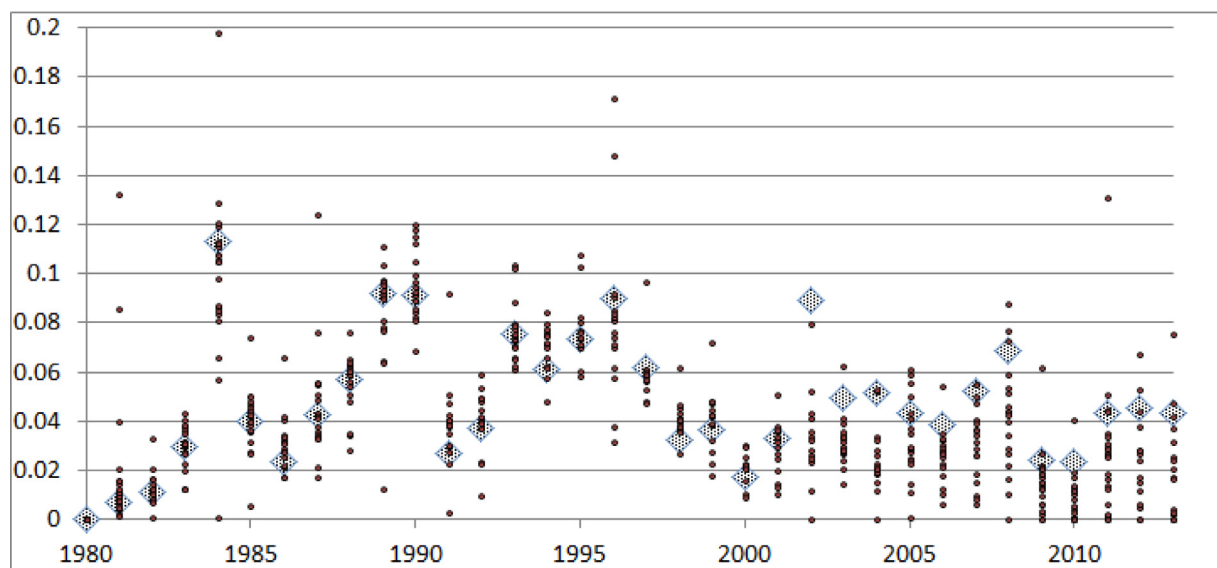
large owners act independently, the threat of exit will be higher. This is consistent with the findings of both [Bharath et al. \(2013\)](#) and [Edmans et al. \(2013\)](#) who show that multiple blockholders strengthen this threat. [Kandel et al. \(2011\)](#) provide further evidence, showing that small investors also govern through the threat of exit because they trade together without agreeing to do so (what they term “unintentional coordination”). If institutions in cliques coordinate, we expect governance via exit to

be less effective when owners belong to cliques. In this section we test this prediction.

5.1. Clique ownership and trading

We first provide evidence on realized exit. As discussed in [Holden and Subrahmanyam \(1992\)](#) and [Foster and Viswanathan \(1996\)](#), if multiple agents receive similar information and cannot credibly commit to cooperate,

(C) Divest



(D) Acquire

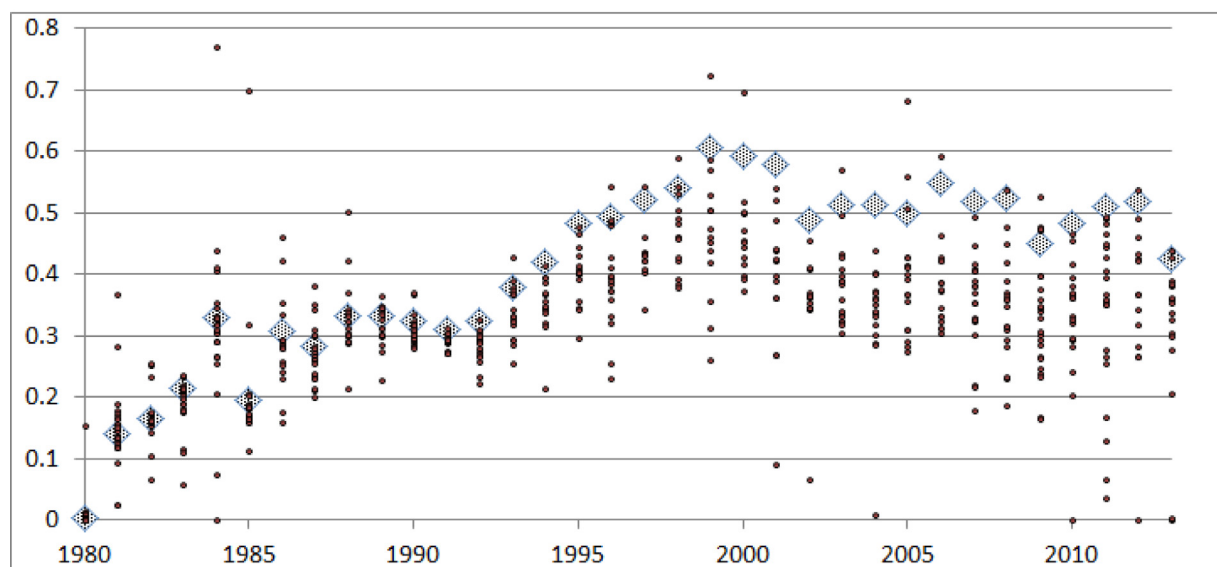


Fig. 3. Continued

this information will quickly be impounded into prices, as the agents trade contemporaneously and aggressively to exploit the value of their information. On the other hand, if the agents can coordinate with each other, then we expect them to sell their positions slowly to minimize price impact.

We use changes in quarterly holdings data to infer trading behavior of institutions and present the results in Table 8. For each institution-stock pair in each quarter, we estimate the change in the position (as a percent of the total market cap of the stock) as a function of the lagged change in the position for that institution-stock pair. We expect institutions in cliques to coordinate their

trades. Therefore they should change their positions more slowly over time (i.e. their trades should be more autocorrelated) in stocks that they co-own with other members of the clique. In column 1–3, we present results consistent with this prediction. We regress the change in ownership (ΔOwn) at time t on the change in ownership at $t - 1$. We also include an interaction with an indicator, *Multiple clique owners*, set to one if the stock in the institution-stock pair is owned by multiple members of the institution's clique, and zero otherwise.

We have no prediction on the overall level of the autocorrelation of trades. This depends on a variety of factors, including when and how information arrives,

Table 7

Specialized cliques and future firm outcomes.

The dependent variables are dividend initiations, Repurchase initiations, Acquisitions, and Divestitures measured at $t + 1$. Dividend, Repurchase, Anti-acquisition, and Divestiture clique ownership measures ownership by cliques in the top quartile of the propensity to own firms with these characteristics at t . Control variables as in Table 4 are included but suppressed for space. All regressions include year and firm effects. Standard errors are clustered by firm with standard errors reported in parenthesis and significance represented according to $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

	(1) Dividend initiation $t+1$	(2) Repurchase initiation $t+1$	(3) Acquisition $t+1$	(4) Divestiture $t+1$
Dividend clique ownership	0.015* (0.01)	0.015 (0.02)	0.014 (0.02)	−0.017* (0.01)
Repurchase clique ownership	−0.008 (0.01)	−0.009 (0.01)	−0.007 (0.01)	−0.005 (0.01)
Anti-acquisition clique ownership	0.004 (0.01)	−0.005 (0.01)	−0.029* (0.01)	−0.022*** (0.01)
Divestiture clique ownership	−0.007 (0.01)	−0.008 (0.01)	−0.013 (0.01)	0.018** (0.01)
Clique ownership	0.012 (0.02)	−0.039 (0.04)	−0.083* (0.04)	−0.018 (0.02)
Institutional ownership	−0.017 (0.02)	0.076** (0.03)	0.131*** (0.04)	0.010 (0.02)
Observations	129,382	104,354	130,000	130,000
Controls	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes
Firm effects	Yes	Yes	Yes	Yes

Table 8

Clique membership and coordinated trading.

This table presents a regression for every institution-stock pair of the institution's change in ownership (ΔOwn_t) on its lagged changes in ownership in the stock. *Multiple clique owners* is equal to one if the stock in the institution-stock pair is owned by other members of the institution's clique in that quarter, and zero otherwise. *Sale* is a dummy variable equal to one if the lagged change in ownership is negative, and zero otherwise. Included fixed effects are indicated at the bottom of the table. Standard errors are clustered by stock quarter and reported in parenthesis. Significance is represented according to $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

	Independent variable: ΔOwn_t					
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{Own}_{t-1} \times \text{Multiple clique owners}$	0.0935*** (0.02)	0.0748*** (0.01)	0.0759*** (0.01)	0.0468** (0.02)	0.0294 (0.02)	0.0305* (0.02)
$\Delta\text{Own}_{t-1} \times \text{Multiple clique owners} \times \text{Sale}_{t-1}$				0.0082 (0.02)	0.0476** (0.02)	0.0465** (0.02)
ΔOwn_{t-1}	−0.2469*** (0.01)	−0.2150*** (0.01)	−0.2129*** (0.01)	−0.3601*** (0.02)	−0.3197*** (0.02)	−0.3177*** (0.02)
Multiple clique owners	0.0004*** (0.00)	−0.0002*** (0.00)	−0.0002*** (0.00)	0.0000 (0.00)	−0.0004*** (0.00)	−0.0004*** (0.00)
Sale_{t-1}				−0.0015*** (0.00)	−0.0017*** (0.00)	−0.0017*** (0.00)
$\text{Sale}_{t-1} \times \Delta\text{Own}_{t-1}$				0.3764*** (0.02)	0.3086*** (0.03)	0.3056*** (0.03)
Multiple clique owners $\times \text{sale}_{t-1}$				0.0007*** (0.00)	0.0009*** (0.00)	0.0009*** (0.00)
Observations	29,599,982	29,599,982	29,548,726	29,599,982	29,599,982	29,548,726
Institution qtr. effects	Yes	No	Yes	Yes	No	Yes
Stock qtr. effects	No	Yes	Yes	No	Yes	Yes

whether institutions sell short when they receive negative information, etc. However, we have a clear prediction that the interaction term $\Delta\text{Own} \times \text{Multiple clique owners}$ should be positive under coordination. We find strong evidence of this; the estimate is both positive and statistically significant. This is consistent with clique members spreading their trades over time when that stock is also owned by other members of the same clique. To interpret the economic effect, note that the main effect is negative. Unconditionally, changes in ownership positions are negatively autocorrelated, i.e., buys on average precede sells. This negative autocorrelation is reduced by roughly 35%

when traded by an institution that co-owns the stock with clique members.

In columns 4–6 we examine if coordinated trading is a function of the direction of trading. On one hand, if two investors can coordinate to minimize the impact of their trades, they should do so no matter the direction of trading. On the other hand, it can be harder to effectively coordinate over positions that the institutions do not yet have. Therefore, there can be a differential effect for exit, where the positions in the clique are clear. We find strong evidence of this. For example, in column 6 we show that the unconditionally negative autocorrelation

shown in columns 1–3 is driven by purchases.²¹ For sales, the autocorrelation for institutional trades in stocks not owned by multiple members of the clique is virtually zero ($-0.3177 + 0.3056$). When the stock is owned by multiple members of a clique, sales by clique members are strongly positively autocorrelated (the sum of all coefficients in the column, 0.0641). This is consistent with clique members exiting positions more slowly when other members of the clique also own the stock.

Importantly, our results hold when we include both institution-quarter effect and stock-quarter effects. We therefore identify a differential effect between trades for the same institution as a function of whether it co-owns the stock within its clique. This ensures that these results are not driven by different investment styles of institutions in cliques or differences in the fundamentals of the stocks they own. Rather, this effect is a function of the co-ownership by clique members and is evidence of coordination in the trading behavior of those institutions.

5.2. Clique ownership and the threat of exit

Our evidence supports the view that clique members unwind positions slowly. We therefore expect governance via the threat of exit to be weaker when investors belong to the same clique. This is difficult to test because the threat of exit is unobservable.

We follow prior literature that links the threat of exit to liquidity (e.g., [Edmans, 2009](#)). Liquidity is good for governance via threat of exit (the sensitivity of firm value to liquidity is high) when multiple, independent, large investors own the firm. [Bharath et al. \(2013\)](#) find evidence of this; the values of firms with multiple blockholders increase more in response to positive liquidity shocks than the values of firms with fewer blockholders. In contrast, we expect the value effect of a positive liquidity shock to be weaker (the sensitivity of firm value to liquidity is low) when the firm is owned by cliques, because their coordination should weaken the threat of exit.

Following [Bharath et al. \(2013\)](#) and [Edmans et al. \(2013\)](#), we examine changes in firm value around decimalization that took place between August 2000 and April 2001 in US stock markets ([Table 9](#)). This shock reduced the minimum tick size from 1/16th of a dollar to one cent, leading to a large decrease in bid/ask spreads. Using this event as a plausibly exogenous shock to liquidity addresses the issue that liquidity and governance can be jointly determined by a firm's unobservable characteristics or that governance might cause changes in liquidity.

First, we confirm the results in [Bharath et al. \(2013\)](#) in our sample. Results in columns 2, 4 and 6 indicate that firm value increased around decimalization more for firms with blockholders relative to those without. The increase in q after decimalization is 0.5 higher for firms with ownership by blocks that is one standard deviation higher. Economically, this effect is similar to that estimated in [Bharath et al. \(2013\)](#). After controlling for this effect of

blockholdings, we also interact decimalization with *Clique ownership* and find that, consistent with predictions from theory, the threat of exit is weaker when clique ownership is high. Statistical significance is at the 1% level, and the economic magnitude is roughly equivalent, but of the opposite sign, to that of ownership by multiple blockholders. Columns 3 through 6 use measures of the concentration of clique ownership, and the findings are similar. These results suggest that, under the same identifying assumptions as [Bharath et al. \(2013\)](#), cliques weaken the power of governance through the threat of exit. Thus, liquidity only improves firm value when there are multiple blockholders who are not in a clique.

An alternative interpretation of our findings is that liquidity negatively affects governance through voice ([Coffee, 1991](#); [Bhide, 1993](#)).²² To provide additional evidence on the relation between cliques and the threat of exit, we examine the characteristics of firms cliques own. We expect that, in equilibrium, investors in cliques will own firms in which the relative power of governance by voice is high, and they should avoid owning firms in which the relative power of governance by exit is low. To examine this, we distinguish firms in which the threat of exit is particularly important from those in which it is not. [Edmans \(2009\)](#) shows that the threat of exit as a governing device is strongest when management assigns a high weight to the short-term stock price.

Following [Edmans et al. \(2017\)](#), we measure managers' short-term concerns using the sensitivity of the value of stock and options awarded to the manager that vest within the following year to the stock price. To calculate this, we obtain the number of equity shares that vest within the next year from ExecuComp. We compute the number of options vesting within the following year as the number of options awarded during year t plus the drop in the number of unvested options from $t - 1$ to t . The delta of stock is one, and we use 0.6 as an approximation for the delta of options ([Jensen and Murphy, 1990](#); [Hall and Lieberman, 1998](#)). Thus, the aggregate delta equals the number of vesting stock plus 0.6 times the number of vesting options. Last, to get the sensitivity over year t of the value of vesting equity for a 100% change in stock price, we multiply the aggregate delta by the stock price at the beginning of year t .²³ To complement this, we also use CEO age, as older managers are closer to retirement and are therefore more likely to make short-term choices as a function of the horizon problem ([Dechow and Sloan, 1991](#); [Antia et al., 2010](#)).²⁴

To examine the relation between coordinated ownership and managers' short-term concerns, we regress

²² [Maug \(1998\)](#) and [Faure-Grimaud and Gromb \(2004\)](#) predict the opposite effect of liquidity on governance through voice. Such a relation should work against our findings on the sensitivity of q to ownership structure.

²³ For these tests we require data from ExecuComp that is available from 2006 forward. Our results are robust to excluding backfilled observations identified in [Gillan et al. \(2017\)](#).

²⁴ We view the results on age as suggestive. Under some assumptions, as in certain models of career concerns, older managers can be less myopic. It's worth noting that our results are similar within the set of CEOs above 60, where career concerns are likely to be small. This is suggestive that the horizon problem is more binding in our sample.

²¹ Because of the growth in the asset management industry, there are more purchases than sales over our sample.

Table 9

Cliques and governance by exit: the effect of decimalization on value.

This table presents a difference-in-difference estimation of the effect of decimalization on the relation between firm value and clique ownership. The dependent variable is Tobin's q as defined in Appendix A. The main variable of interest is the interaction of *Decimalization* and one of the three measures of ownership by cliques: *Clique ownership*, *Clique Herfindahl*, and *Clique own.—top 1*. This regression is estimated on years 2000 and 2002 (2001 is the year of treatment and is excluded). Firm-fixed effects are included. Standard errors are clustered by firm with standard errors reported in parenthesis and significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1) q	(2) q	(3) q	(4) q	(5) q	(6) q
Clique ownership _{$t-1$}	1.476 (1.12)	2.294** (1.12)				
Decimalization × Clique ownership _{$t-1$}	−0.687*** (0.19)	−1.842*** (0.28)				
Clique Herfindahl _{$t-1$}			2.783*** (0.71)	4.845*** (1.12)		
Decimalization × Clique Herfindahl _{$t-1$}			−1.040** (0.50)	−4.291*** (1.04)		
Cliques own.—top 1 _{$t-1$}					2.300*** (0.63)	4.199*** (0.79)
Decimalization × Cliques own.—top 1 _{$t-1$}					−0.757** (0.38)	−4.456*** (0.83)
Ownership by blocks _{$t-1$}	0.557 (0.76)	−1.360 (0.83)	−0.660 (0.76)	−2.021** (0.87)	−0.206 (0.83)	−2.248** (0.96)
Decimalization × Ownership by blocks _{$t-1$}		2.686*** (0.39)		2.181*** (0.48)		3.396*** (0.57)
Decimalization	−0.196** (0.09)	−0.176* (0.09)	−0.387*** (0.06)	−0.503*** (0.07)	−0.352*** (0.09)	−0.225** (0.09)
Ln(Market cap) _{$t-1$}	−0.486*** (0.09)	−0.462*** (0.09)	−0.495*** (0.09)	−0.484*** (0.09)	−0.503*** (0.09)	−0.492*** (0.09)
Number of blockholders _{$t-1$}	−0.058 (0.07)	−0.033 (0.06)	0.017 (0.07)	0.035 (0.07)	−0.023 (0.07)	0.001 (0.07)
Book leverage _{$t-1$}	−0.067 (1.08)	0.001 (1.07)	−0.109 (1.08)	−0.075 (1.07)	0.019 (1.09)	0.100 (1.09)
Inst. ownership _{$t-1$}	−1.807* (1.06)	−1.697 (1.05)	−1.459*** (0.54)	−1.433*** (0.54)	−1.361*** (0.52)	−1.166** (0.53)
Annual stock return _{$t-1$}	0.353*** (0.05)	0.338*** (0.05)	0.355*** (0.06)	0.347*** (0.05)	0.354*** (0.06)	0.344*** (0.05)
CapEx _{$t-1$}	−0.000 (0.00)	0.000 (0.00)	−0.000 (0.00)	−0.000 (0.00)	−0.000 (0.00)	−0.000 (0.00)
Dividend payer _{$t-1$}	0.308* (0.19)	0.306 (0.19)	0.314* (0.18)	0.312* (0.19)	0.302 (0.19)	0.284 (0.19)
Observations	7765	7765	7765	7765	7736	7736
Firm effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.127	0.136	0.126	0.130	0.126	0.133

Clique ownership on the sensitivity of vesting equity (and CEO age) and firm controls. We present this analysis in Table 10. Results show that the presence of coordinated investors is weakest among firms with managers that have short-term concerns. A one standard deviation increase in the sensitivity of the value of vesting stock and options is associated with about 1% less of the total firm owned by cliques. Similarly, a manager that is ten years older is associated with 0.4% less ownership by cliques. The magnitude is similar across all measures of clique ownership and is statistically significant at the 10% level, at minimum, in all specifications. Our results are consistent with ownership endogenously adjusting such that in equilibrium investors hold firms in which their marginal productivity of governance is highest.

6. Robustness

6.1. Alternative explanations

An alternative interpretation is that clique members are not coordinating, rather they are acting independently but

give the appearance of coordinating, e.g., vote together, because they have independently acquired correlated information. This is a version of the classic reflection problem as in Manski (1993). Activist investor Phillip Goldstein, in the context of investor coordination famously said, “If you're going to a Grateful Dead concert, you're going to find a lot of Grateful Dead fans. They're not a group. They just like the same music.”

It is possible that our measure of cliques captures this same idea. Three of our prior tests help rule this out. First, there is no reason that investors acting independently would be more likely to disclose joint activism. Therefore, the fact that clique members are more likely to file joint 13Ds is inconsistent with the correlated information alternative. Second, institutions in cliques sell out of positions more slowly when other clique members also own the stock. This is also predicted by coordination and not correlated independent actions. Finally, if clique members are not coordinating but simply have correlated information, then the threat of exit should be stronger, not weaker, in the presence of clique owners.

Table 10

Clique ownership and managers' short-term concerns.

This table presents a regression of the determinants of *Clique ownership* as a function of managers' short-term concerns. *Vesting equity sensitivity* is the measure of managerial myopia as defined in Edmans et al. (2017). *CEO age* is the age of the CEO (measured per 10 years). Year-fixed effects are included. Standard errors are clustered by firm with standard errors reported in parenthesis and significance represented according to * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1) Clique ownership	(2) Clique ownership	(3) Clique Herfindahl	(4) Clique Herfindahl	(5) Top cliques own.	(6) Top cliques own.
Vesting equity sensitivity _{t-1}	-0.098*** (0.03)		-0.031*** (0.01)		-0.046** (0.02)	
CEO age		-0.004* (0.00)		-0.002*** (0.00)		-0.002** (0.00)
Ln(Market cap) _{t-1}	0.036*** (0.00)	0.036*** (0.00)	0.009*** (0.00)	0.007*** (0.00)	0.014*** (0.00)	0.012*** (0.00)
Ln(Market to book) _{t-1}	0.003 (0.00)	0.006** (0.00)	0.003* (0.00)	0.002*** (0.00)	0.005* (0.00)	0.002 (0.00)
Dividend payer _{t-1}	-0.022*** (0.01)	-0.021*** (0.00)	-0.004* (0.00)	-0.002** (0.00)	-0.004 (0.00)	-0.001 (0.00)
Number of blockholders _{t-1}	0.021*** (0.00)	0.023*** (0.00)	-0.007* (0.00)	-0.005*** (0.00)	-0.007* (0.00)	-0.008*** (0.00)
Block ownership _{t-1}	0.337*** (0.04)	0.426*** (0.03)	0.304*** (0.05)	0.278*** (0.02)	0.378*** (0.05)	0.414*** (0.02)
Annual stock return	0.008** (0.00)	0.017*** (0.00)	-0.001 (0.00)	0.001 (0.00)	-0.003* (0.00)	0.001 (0.00)
Observations	6234	21,420	6234	21,420	6233	21,418
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.558	0.572	0.519	0.535	0.487	0.496

To provide one more piece of evidence on coordination among clique members, we examine if ownership by cliques substitutes for other mechanisms by which owners might coordinate. Crane and Koch (2018) argue that private securities class action serves as coordination mechanism for owners to discipline management. Using a court decision that provides a plausibly exogenous reduction in the ability of investors to file class action litigation, they show that ownership concentration increases as a response to an increase in the free-riding problem. We show that this effect is weaker when firms are owned by cliques.

In 1999, the Ninth Circuit Court of Appeals issue a decision in *Re: Silicon Graphics* that set a more stringent hurdle for filing securities class action litigation. This ruling disproportionately affected firms headquartered in the Ninth Circuit. See Crane and Koch (2018) for a detailed discussion of the ruling and the exogenous nature of the decision. In online Internet Appendix Table IA-1.4, we regress institutional ownership on a *Treatment* indicator equal to one if the firm is subject to this ruling. We find that treated firms without clique ownership see a statistically significant 8% increase in institutional ownership (and similar percent increases in ownership concentration and the number of large shareholders) following the loss of the coordination mechanism. However, this effect is significantly weaker when the firm is owned by cliques. This is consistent with the view that ownership cliques are less reliant on alternative formal coordination mechanisms. These results suggest cliques are capturing coordination and are generally inconsistent with the alternative explanation that clique ownership is picking up correlated information.

6.2. Alternative measures of coordination

Institutions that are working together can purposefully avoid owning large stakes individually. We recompute

Clique ownership, excluding positions larger than 5%.²⁵ This ensures that our findings are not driven by the large positions that actually define the network, but rather the smaller positions that cliques own.²⁶

Further, we also repeat our analysis using a cluster coefficient instead of our measure of cliques. A cluster coefficient is an institution level measure designed to reflect how close an institution's connected peers are to being a clique. Cluster coefficients are easy to compute and easily adaptable to several different definitions of connections between institutions. The primary drawback of a cluster coefficient is that, while it reflects the likelihood that an investor belongs to a clique, it does not identify the clique to which the investor belongs. We repeat our core analysis using a cluster coefficient and a variety of definitions for connections. Consistent with our main specifications, we find that clustered ownership facilitates voice and mitigates the threat of exit. The results using the cluster coefficient are presented throughout online Internet Appendix IA-3.

7. Conclusion

Using a novel measure of ownership coordination derived from the intersection of network theory and economics, we show that firms with high levels of ownership by cliques of institutional investors experience more direct intervention in the form of votes against bad management proposals. Evidence from a plausibly exogenous shock to the network of institutional investors suggest that this relationship between coordination among institutional

²⁵ We still use the $\geq 5\%$ positions to define the network connections, but then exclude these positions from all other analysis.

²⁶ These results are in online Internet Appendix IA-4.

investors and governance is causal. Furthermore, we find that institutions are more likely to file disclosures of joint activism with the SEC when they belong to cliques. We also find that the presence of certain ownership cliques predicts future corporate outcomes traditionally associated with activism.

While these ownership cliques coordinate to improve governance via voice, they also coordinate to minimize the price impact of their trades. Consistent with this, we find that governance via threat of exit is weaker in the presence of ownership cliques. This is in line with Edmans and Manso (2011) who show theoretically that free riding (coordinating) among owners strengthens (weakens) the threat of exit.

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